THE SOCIETY THAT SELECTS US

Fertility, Personality, and the Human Futures We Build Without Realizing It

THE SOCIETY THAT SELECTS US

Fertility, Personality, and the Human Futures
We Build Without Realizing It

A scientific and philosophical investigation of selection, culture, urbanization, and society's future

NOTE TO THE READER

This book is an essay grounded in scientific literature, not an argument for eugenics or a moral classification of populations. Asking whether societies create selective pressures is scientifically legitimate; the answer quickly goes wrong, however, if we turn statistical averages into the essences of peoples, confuse cultural correlation with genetics, or treat psychiatric diagnoses as evolutionary identities.

We will distinguish four phenomena under “selection”: genetic selection, when heritable variation covaries with reproductive success; cultural selection, when ideas and norms spread differentially; social filtering, when institutions attract or exclude certain profiles without changing their genetic frequency; and demographic selection, when subgroups with different fertility rates change their relative proportions. These mechanisms can combine, but they are not synonyms.

When comparing China, India, Europe, Indigenous communities, or urban and rural populations, our focus is on institutions, ecology, norms, migration, and measurement. The “age of a civilization” is not a well-defined biological variable. Individual variation within any population is substantial, and comparisons of cultural averages require caution.

The question we cannot avoid

Some questions sound dangerous precisely because they were poorly framed in the past. Here is one: does society change only our culture, or does the way it distributes opportunities to form couples and have children begin to change us genetically as well?

Common intuition places natural selection in prehistory. Medicine has reduced mortality, the state has reduced internal violence, and technology has sheltered us from many environmental whims. But evolution does not require tigers, famine, or epidemics. It requires only heritable variation and systematic differences in reproduction. In a society where one group has an average of 0.9 children and another 3.2, something inevitably changes between generations. The question is what changes: genes, norms, religion, social status, geography, everything simultaneously, or almost nothing of biological significance.

This question has become more urgent because global demography has turned a corner. The United Nations estimates that the global population, 8.2 billion in 2024, will grow to approximately 10.3 billion in the 2080s and then begin to decline. Global fertility was 2.25 children per woman, and more than half of countries and territories were already below the approximate replacement threshold of 2.1. Another model, from the Global Burden of Disease study, projects global fertility of 1.83 in 2050 and 1.59 in 2100 in its reference scenario. These are projections, not prophecies, but the direction is serious enough to shift the question from "how much will we grow?" to "what kinds of societies can continue to reproduce?" (UN DESA, 2024; GBD 2021 Fertility and Forecasting Collaborators, 2024).

The book will not argue that "we are all heading for extinction." On the timescale of the twenty-first century, that would be an exaggeration. It will argue something more interesting: modernity has created, probably unintentionally, a new reproductive environment. Within it, economic success, education, urbanization, autonomy, housing costs, time at work, relationship norms, and access to kinship networks influence not only how we live but also who ends up having descendants.

Here lies the paradox. A society can reward a trait professionally while penalizing it reproductively. It can value education while making the years education requires incompatible with starting a family early. It can select professionally for mobility and independence while destroying precisely the networks of grandparents, relatives, and neighbors that make raising children manageable. Institutions are not merely stages on which people act out their biology. They become part of the evolutionary environment.

Yet culture moves far faster than genes. This is the key to the entire book. Norms can reverse an association within decades; migration can reshape a population within a generation; housing policy can change the age at first birth without altering a single genome. Any account that jumps directly from "fewer children are born in cities" to "cities genetically select human type X" misses the very mechanisms that matter.

We will therefore walk a narrow ridge: going far enough to formulate uncomfortable hypotheses, yet remaining rigorous enough to recognize where the evidence ends. The result is not a book about genetic purity. It is a book about institutional freedom: what kinds of social environments allow different human types to exist, cooperate, and have a future.

WORKING THESIS CENTRAL HYPOTHESIS

Societies are environments of selection, but today they select norms, life strategies, and demographic compositions far faster than genes. Genetic effects exist and can be measured, but they are small, polygenic, context-dependent, and often mediated through education, age at first birth, migration, and access to family life. Our biological future cannot be separated from institutional design.

CONTENTS

PART I

SELECTION HAS NOT STOPPED

Evolution continues even when mortality declines. Only the criteria change.

CHAPTER 1

Evolution did not get the news about modernity

Natural selection after natural selection stopped looking natural

In the modern imagination, human evolution has an unspoken ending: the moment we invented medicine, sanitation, antibiotics, and the welfare state. Before that, nature selected. Afterward, culture supposedly suspended the game. Biologically, this picture is wrong. Natural selection is synonymous neither with early death nor with “survival of the strongest.” It arises wherever heritable differences covary with the number of descendants who themselves go on to reproduce.

If almost everyone survives to reproductive age, selection does not disappear; its center of gravity shifts from survival toward mating, reproductive timing, and the number of children. In such an environment, a difference of one child on average per adult can matter more evolutionarily than a small difference in mortality.

This does not mean that every fertility difference is genetic. For a trait to respond to genetic selection, the relevant variation must be heritable to some extent. Personality partly satisfies this condition. A meta-analysis of behavioral genetic studies estimated that approximately 40% of individual variation in personality is associated with genetic differences, with the remainder attributed to environment and measurement error; estimates are higher in twin studies than in adoption and family studies (Vukasović & Bratko, 2015). But “40% heritable” does not mean “40% predetermined” and says nothing about differences between peoples. Heritability is a statistical property of variation within a population in a given environment.

An equation without unnecessary mathematics

At its simplest, evolutionary change can be imagined as follows: if a heritable characteristic is systematically more prevalent among people who have more children, the average of that characteristic may increase in the next generation. Quantitative genetics offers precise formalizations, including the Price equation, but the intuition is a covariance: trait x reproductive success.

The problem is that people do not reproduce in a laboratory. Between a trait and a child stand university, rent, the labor market, dating culture, religion, age at marriage, access to contraception, migration, health, family policy, and the partner's preferences. Society introduces dozens of intervening variables. An evolutionary effect can be real even when its proximate cause is almost entirely institutional.

A well-known example comes from Iceland. Kong and colleagues showed that a polygenic score associated with educational attainment predicted later reproduction and, on average, fewer children. Among cohorts born between 1910 and 1990, the average score declined by approximately 0.010 standard deviations per decade. The authors interpreted the result as selection against education-associated variants in that context (Kong et al., 2017). This is an important finding, but it is not a verdict on “declining intelligence.” Polygenic scores capture a fraction of variation, depend on the population in which they were developed, and are not equivalent to a single moral or cognitive trait.

More importantly, the environment can reverse selection. A recent study of 22,636 people in 17 countries found an average positive association between conscientiousness and number of children, a negative association for high openness, but small effects and different signs across countries for extraversion, neuroticism, and openness (Sanz-García et al., 2025). There is therefore no single “reproductive profile of modernity.” There are reproductive ecologies.

Cultural evolution races ahead

A custom can become dominant within a generation; a rare allele usually takes much longer. For this reason, most change we observe in the twenty-first century will be cultural and demographic before it is genetic. High-fertility communities can rapidly increase their share through vertical transmission of religion or family norms, without the difference having a genetic basis. A city can change its psychological profile because it attracts more open or proactive migrants, not because it genetically “produces” them.

Separating these mechanisms does not diminish the evolutionary question. On the contrary, it makes it more serious. Modern society has not stopped selection. It has moved it into a labyrinth of institutions.

EVIDENCE VERDICT STRONG, WITH LIMITATIONS

Detectable biological selection exists in modern human populations, including on polygenic traits associated with reproduction and education. But the effects are small and context-dependent. A genetic future cannot be inferred directly from fertility differences.

CHAPTER 2

The great demographic paradox

We are not disappearing tomorrow. But societies that do not reproduce outsource their future.

In 1968, the dominant fear was a population explosion. In many wealthy societies of 2026, the anxiety has reversed. South Korea, Japan, Italy, Spain, and much of Europe experience persistently below-replacement fertility. China has moved from its one-child policy to attempts to stimulate births. For the first time in a long while, the demographic future is imagined not as crowding but as emptiness.

Globally, however, the word “disappearance” is premature. The UN's World Population Prospects 2024 estimates growth from 8.2 billion in 2024 to around 10.3 billion in the 2080s, followed by a slow decline toward 10.2 billion in 2100. Global fertility was estimated at 2.25. The species is therefore not in a terminal demographic spiral this century (UN DESA, 2024).

But the global aggregate hides two worlds. Some regions are aging and shrinking; others remain young and growing. The GBD 2021 model projects global fertility of 1.59 in 2100 in its reference scenario, with births increasingly concentrated in sub-Saharan Africa. The UN and GBD use different models, and the differences between their projections are precisely why they must not be read as destinies. What is robust is the shift in demographic regime: smaller families, later births, more childhoods without siblings, more childless people, and a massive geographical redistribution of births.

The replacement threshold is not a prophecy

The approximate figure of 2.1 is often treated as a magic wall. It is not. A population with fertility below this level can continue to grow for a time because of age structure and migration. A population above the threshold can decline under certain conditions of mortality and emigration. Yet if fertility remains well below replacement for enough generations, without immigration, the arithmetic becomes unforgiving: each generation is smaller.

This introduces a political paradox. Societies with very high productivity and human capital can sustain aging populations for a long time. Automation can reduce the need for workers. Migration can compensate for local decline. But none of these solutions fully answers the evolutionary question: who passes on norms, life strategies, and, to a lesser extent, heritable predispositions?

A city that does not reproduce its inhabitants can remain vibrant by continually importing young people. Demographically, it functions as a basin of attraction. Culturally, it transforms migrants. Evolutionarily, it can be a “sink”: an environment people enter, have fewer children than they would in other settings, and are replaced by new waves of mobility. This image does not condemn the city; it shows how urban density can be sustained by an external demographic metabolism.

Cultures disappear before people do

A population does not have to reach zero for something to be lost. Languages, kinship networks, crafts, religious denominations, social classes, and ways of life can shrink until they can no longer reproduce institutionally. A society can survive numerically yet become a different society.

This is the first philosophical shift the book proposes. The useful question is not “will humanity disappear?” but “which social configurations can reproduce without coercion, and which traits, norms, and institutions become more common when some configurations do not reproduce?”

For a civilization, fertility is not merely a private statistic. It is the mechanism through which the present receives a vote in the future.

ESSENTIAL DISTINCTION VERY STRONG

Local demographic decline, compositional change, and species extinction are different phenomena. Current data strongly support the first two in many regions; they do not support imminent global extinction.

CHAPTER 3

Who has children?

Fertility is where private values meet social architecture.

In societies with effective contraception, the number of children is less constrained by biology and more mediated by decisions. Paradoxically, this can make selection more dependent on psychology and institutions, not less.

Several robust regularities exist. In Europe, fertility is generally lower in urban than in rural regions. For 2022, Eurostat reported a total fertility rate of 1.39 in the EU's urban regions, 1.49 in intermediate regions, and 1.57 in rural regions; in 19 of 23 states with comparable data, urban fertility was lower than rural fertility (Eurostat, 2024). Using 1,328 NUTS 3 regions in 28 European countries, Riederer and Beaujouan showed that births also occur later in cities; much of the gradient diminishes when women's education, wealth, high-tech employment, family norms, and population composition are included (Riederer & Beaujouan, 2024).

The “city,” then, is not a sterilizing substance. It is a package: housing costs, opportunities, prolonged education, larger marriage markets, mobility, distance from relatives, competitive careers, and more individualized norms. Once we unpack it, the urban effect becomes more intelligible.

Personality enters the equation but does not dictate the outcome

Studies of the Big Five and fertility are fascinating precisely because they offer no simple story. In a World Values Survey analysis across 17 countries, conscientiousness had, on average, a positive association with number of children. Very high openness was associated with fewer children, but effects were small, and coefficients for extraversion, neuroticism, and openness changed from positive to negative across countries (Sanz-García et al., 2025).

This is exactly what we would expect if institutions altered a trait's “reproductive value.” Conscientiousness can facilitate organizing a family in a state with good services and be consumed by a hypercompetitive career in another. Openness can delay settling down in an environment that rewards mobility and experimentation, yet increase the chances of finding compatible partners in a diverse setting. There is no theoretical reason why the same gradient should be universal.

Historical and local studies also confirm this variability. In some American cohorts, openness and, among women, conscientiousness were associated with lower fertility, while extraversion and agreeableness showed some positive associations. In high-fertility populations with subsistence economies, such as certain rural communities in Senegal, the relationships between personality, status, polygyny, and reproductive success look different. Context changes the mechanism.

The marginal child

Selection concerns not only who has children and who does not, but also the marginal child: the difference between one and two, or two and three. Having one child may be compatible with almost any career path; having three may require a larger home, stable income, available grandparents, flexible work, and a social norm that does not treat a large family as eccentric.

Policies and infrastructure therefore have evolutionary potential even when they pursue no biological goal. A planning rule that makes a three-bedroom apartment near work impossible may affect the descendants of certain life strategies more than a pronatalist campaign. A good after-school program can change the calculation for a second child. Remote work can redraw the geography of family support.

Modern selection often enters through the ordinary door of logistics.

CHAPTER 4

How institutions turn preferences into descendants

Intervening mechanisms matter more than labels.

Suppose two people express the same desire to have two children. One lives in a community with affordable housing, nearby grandparents, predictable schedules, and a culture that includes parents in social life. The other lives in a metropolis where career advancement requires mobility, rents rise faster than wages, commuting consumes two hours a day, and close friends move frequently. Their preference is the same. Their probability of realizing it is not.

Institutions turn preferences into descendants through at least five filters.

1. The partner market

The size and structure of the social network determine who meets whom. Universities, professions, and dating platforms produce assortative mating: people similar in education, values, or lifestyle pair up more often than chance would predict. This phenomenon can amplify cultural and genetic differences, but its effect depends on the trait being examined.

2. Timing

Age at first birth is one of the strongest bridges between education, career, and completed fertility. Kong's Icelandic study demonstrates precisely this partial mediation: the polygenic score for education was associated with later reproduction, which explained a substantial part of the smaller number of children. In evolutionary terms, time is a resource.

3. Opportunity cost

The more a society concentrates human capital in the years from 25 to 40, the more expensive the conflict between career and parenthood becomes. It is not enough here to say “people choose careers.” The choice is produced by an architecture of rewards: promotions, grants, tenure, start-ups, bonuses, relocations. Some systems place the biological fertility window in direct competition with the period of greatest professional acceleration.

4. Care infrastructure

Homo sapiens is a species that raises children cooperatively. Grandparents, relatives, older siblings, and the community have long mattered. Urban nuclear families sometimes try to compress this network into a couple of two adults, each working full-time. It is unsurprising that a second or third child becomes costly.

5. Prestige

Behaviors reproduce culturally through imitation. If high-status adults have few or no children, and success is visible precisely through freedom from family obligations, a cultural feedback loop can arise. Conversely, communities where parenthood provides status and social networks can sustain higher fertility without massive financial incentives.

These filters suggest a better model than “society selects personality X.” Society selects compatibilities between traits and institutions. A highly open, mobile person may reproduce successfully where community and services reduce the cost of mobility; that person may have low fertility where every geographical move breaks the support network.

An institution is evolutionary when it systematically changes who manages to turn the desire for a family into an actual family.

This formulation has an important ethical consequence. We do not need to optimize populations by genotype or psychological profile. Instead, we can design institutions that reduce arbitrary distortions: making it unnecessary to choose between education and children, autonomy and community, or city and family. In this sense, a pluralistic society allows more life strategies to reproduce.

PART II

PERSONALITY IS AN ECOLOGY

People differ, but how we measure and combine those differences depends on the world they inhabit.

CHAPTER 5

How much of personality is inherited?

Heritability is real. Determinism is a category error.

To discuss genetic selection on personality, we must cross a conceptual minefield: heritability. It is one of the statistics most easily turned into a metaphysics.

Vukasović and Bratko's meta-analysis, synthesizing more than a hundred thousand participants, estimates average personality heritability at around 0.40. In twin studies, the average estimate was 0.47; in family and adoption studies, 0.22. The difference shows that methodology matters and that the figure is not a fixed truth about the species.

Heritability answers this question: within a particular population, under its environmental conditions, how much observed variation between individuals is statistically associated with genetic variation? It does not answer: how much of an individual's personality is “genetic”? Still less does it answer: are the average differences between two cultures genetic?

A classic example helps. If all children in a population receive almost identical nutrition, genetic variation may explain a large proportion of variation in height. If we change nutrition across the entire population, average height may change dramatically without genes having changed. A trait can be highly heritable yet highly sensitive to the environment.

Personality is not a set of five genes

Personality traits are polygenic and development-dependent. Thousands of variants can contribute extremely small effects, while early experiences, relationships, stress, illness, status, culture, and life events shape their expression. Even personality stability increases with age without becoming rigidity.

Studies of international mobility provide an instructive picture. Zimmermann and Neyer followed students before, during, and after periods of study abroad. They found both selection—certain personality profiles predicted who would leave—and socialization: international experience was associated with subsequent changes in openness, agreeableness, and neuroticism, partly mediated by changes in support networks (Zimmermann & Neyer, 2013). Thus, a global city's population can look psychologically different through two simultaneous mechanisms: it attracts a certain kind of person and then changes that person.

Genes can respond to culture

Gene-culture coevolution is not a metaphor. Classic examples such as lactase persistence show that cultural practices can create biological selective pressures. But the leap from milk to personality is enormous. For complex behavioral traits, genetic architecture is diffuse, environmental influence is strong, and the meaning of the same predisposition changes with institutions.

This makes the future less deterministic, not less evolutionary. If a norm or policy can rapidly change the relationship between a predisposition and reproductive success, social design is part of biological causality. We do not have a “genetic program” pushing us toward a single destination. We have a distribution of predispositions encountering environments we construct.

READING RULE FUNDAMENTAL CONCEPT

A trait can be heritable within each population while the average difference between two populations is nevertheless entirely environmental. Heritability does not automatically explain differences between groups.

CHAPTER 6

The Big Five and the problem of a universal ruler

Sometimes the difference between cultures lies in people. Sometimes it lies in the instrument.

A book about cultural differences in personality could be very easy to write and very easy to get wrong. We could take average scores for extraversion, agreeableness, conscientiousness, neuroticism, and openness from dozens of countries, then say: this culture is more disciplined, that one more anxious, another more open. The result would be elegant charts and fragile conclusions.

The problem is called measurement invariance. To compare two populations' averages, an instrument must measure the same construct on the same scale. A major review published in 2026 examined 233 publications on personality assessment across cultures and found no instrument with evidence of full cross-cultural scalar invariance. NEO-PI-R and IPIP-NEO-120 had some of the strongest evidence, but the authors explicitly caution against simple inferences about mean differences between cultures (Sheppard et al., 2026).

This does not mean all comparisons are impossible. It means we must scrutinize the instrument as closely as the phenomenon.

The stereotype is not the average

A remarkable 2005 study compared stereotypes of “national character” with measured personality across 49 cultures. The stereotypes were consensual but barely correlated with averages from self-reports and observer ratings: the median correlation was approximately 0.04 (Terracciano et al., 2005). People may strongly agree about “what Italians are like,” “what Japanese people are like,” or “what Romanians are like,” yet that consensus may not be a good psychometric description.

This is a warning for the chapters on China and India. We will discuss statistical differences and cultural structures without turning civilizations into persons.

The Big Five can be real without being completely universal

The Big Five is an extraordinarily useful model in industrialized populations and has been replicated in many languages. But “five” may partly reflect the kinds of social niches people inhabit. Among the Tsimane, a population of horticulturalists and foragers in the Bolivian Amazon, Gurven and colleagues did not robustly reproduce the Big Five structure in either self-reports or spouses' ratings. The data instead suggested two broad dimensions, close to prosociality and industriousness (Gurven et al., 2013).

Smaldino and colleagues subsequently proposed the niche diversity hypothesis: complex societies offer many viable roles and combinations of behaviors; this diversity reduces covariance between traits and allows more statistical dimensions to emerge. A later test across 115 nations and 685,089 people found that niche diversity was associated with lower covariance between traits and greater personality dimensionality (Smaldino et al., 2019; Lukaszewski et al., 2022).

Here emerges one of the book's strongest insights: a complex society may create the impression of greater psychological diversity not because mutations have suddenly created new people, but because more stable ways of being human exist. A village where almost all adults must simultaneously be workers, relatives, caregivers, and negotiators may favor coherent behavioral packages. A metropolis allows the brilliant, introverted, disorganized specialist to survive professionally alongside the hyper-social manager, nomadic artist, and highly conscientious accountant.

Institutional complexity can therefore preserve diversity. This idea will matter enormously when we discuss the tribal future versus the megacity.

CHAPTER 7

The village, the state, and the bureaucrat

Institutions have no personality, but they reward personalities.

A community of a hundred people and a state of a hundred million are not the same problem at different scales. They require different social technologies.

In small groups, reputation is dense. A broken promise travels quickly. Kinship and repeated relationships substitute for contracts. Ambiguity can be resolved through personal knowledge. In large societies, people must cooperate with strangers. Documents, abstract rules, procedures, audits, registers, and bureaucrats emerge here.

It is tempting to say that old civilizations “genetically selected” obedience, patience, or conformity. Evidence for such a claim is weak. What we can say more confidently is that institutions culturally and professionally select behaviors compatible with them. Examinations favor certain competencies; armies favor discipline; entrepreneurial markets reward initiative and risk tolerance; administrations reward predictability and the ability to work with abstract rules.

These systems can have reproductive effects only if success within them is systematically connected to family formation and number of children. In some eras, bureaucratic status or property ownership increased fertility. In others, lengthy education and administrative-professional careers delay it. The same type of institution can reverse the direction of selection over time.

Tight cultures and loose cultures

Michele Gelfand and colleagues' classic study of 33 nations introduced a useful axis: tightness-looseness. “Tight” cultures have strong norms and low tolerance for deviance; “loose” cultures have weaker norms and greater tolerance. In their model, tightness was associated with historical and ecological threats, such as high density, resource scarcity, territorial conflict, and disease, together with more restrictive institutions and a greater psychological need for structure (Gelfand et al., 2011).

We should not read this axis as a scale from bad to good. In an environment with substantial epidemiological or military risk, conformity can be collectively adaptive. In a stable environment, tolerance for deviance can produce innovation and exploration. The future's problem is that technology can simultaneously combine global threats with enormous individual possibilities. What degree of tightness will be functional?

Bureaucracy as a niche

Modern bureaucracies do something subtle: they outsource to procedures part of the trust that, in a village, resides in relationships. This enables cooperation on an extraordinary scale. But it can also separate social competence from consequences. Someone can make decisions affecting thousands of strangers without direct reputational feedback. In this sense, the institution changes the ecology of traits such as empathy, conformity, dominance, or scrupulousness.

The empirical question is not the sensationalist “does bureaucracy produce psychopaths?” but “which combinations of traits do different organizational architectures reward, eliminate, or protect?” A system with auditing, transparency, and collective decision-making can limit interpersonal exploitation. An opaque, highly hierarchical system can turn lack of remorse into a local advantage. An institution can alter a strategy's cultural fitness without immediately changing the frequency of a genetic predisposition.

Here political philosophy meets evolution: constitutions are also mechanisms of behavioral ecology.

CHAPTER 8

China: one civilization, many psychologies

Historical age is not an explanation. Institutions, ecology, and kinship may be.

China is the ideal case for showing how quickly the notion of a "civilization's personality" becomes toxic. More than a billion people, hundreds of millions of internal migrants, enormous differences between coast and interior, urban and rural areas, generations educated before and after liberalization: any national average is a brutal compression.

Yet China offers valuable natural experiments precisely because many historical and ecological differences exist within a large population with partially shared language, state, and history.

Rice and wheat: elegant hypothesis, disputed evidence

In 2014, Thomas Talhelm and colleagues tested 1,162 Han Chinese people across six sites and reported that historical rice-growing regions in the south showed greater interdependence and holistic thinking than northern wheat-growing regions. The explanation was the intensive cooperation required for irrigation and rice cultivation (Talhelm et al., 2014).

The story is seductive: agricultural technology -> local institutions -> psychology. But a 2015 critique showed that important results weakened or disappeared after corrections for sampling bias, measurement error, and model specification, while alternative collectivism data failed to reproduce the expected relationship (Ruan, Xie & Zhang, 2015). The lesson is not that agriculture does not matter. It is that beautiful cultural theories must be treated as falsifiable theories, not explanatory myths.

The clan as institutional memory

A 2026 paper examined the persistence of clan institutions and contemporary collectivism using three types of data: questionnaires from 6,700 participants, indicators derived from approximately 300,000 Weibo users across numerous cities, and historical data covering 816 districts and counties. Presence and involvement in clan structures were associated with more collectivist orientations (Ji, Liu & Zhu, 2026). The study does not demonstrate genetic causality; rather, it offers an example of institutional memory: historical forms of organization can leave psychological and linguistic traces long after the economy has changed.

Indigenous personality models and what the Big Five misses

Asian research has also developed indigenous instruments. The Chinese Personality Assessment Inventory identified dimensions such as Interpersonal Relatedness that do not emerge as clearly in imported Western inventories. This does not mean Chinese people have a “relatedness gene”; rather, a taxonomy constructed within one culture may notice behavioral regularities that another compresses or misses (Cheung et al., 2003).

Demography changes the problem again

Today China is also a laboratory of very low fertility. Urban-rural differences remain important, although they are changing. A state once concerned with limiting births now worries about the cost of parenthood. This reversal within a single generation should cure us of determinism. The same populations can move from pronatalist norms to very low fertility without relevant genetic changes.

The Chinese case shows that “ancient civilization” is not a cause. What matters is the persistence of institutions, ecologies requiring cooperation, labor markets, urbanization, family norms, and the ways these reorganize. Age is a chronology; the mechanism must be sought elsewhere.

CHAPTER 9

India: the impossibility of a single average

When a political unit encompasses hundreds of social ecologies, the average can become fiction.

If China is difficult to reduce to a personality, India makes the exercise almost absurd. Different languages, religions, castes, classes, regions, kinship systems, urban densities, and colonial histories coexist within a single state. Speaking of “Indian personality” without precisely defining the sample is closer to literature than science.

Local studies are useful for exactly this reason. A study in Karnataka, southern India, applied an English or Kannada version of the BFI-2-S to 400 community participants. Factor analysis produced four- and five-factor solutions that did not map neatly onto the Big Five; the authors labeled the resulting five factors Social Effectiveness, Interpersonal Ability, Altruism, Emotional Instability, and Innovativeness (Thomas et al., 2023). This is an interesting result, but the sample is too small and local to become “India's model.”

What it demonstrates is something else: importing a psychological ruler into another context may produce a different geometry.

Beyond East versus West

A multinational study led by Vivian Vignoles tested models of selfhood in two samples: 2,924 students from 16 nations and 7,279 adults from 55 cultural groups across 33 nations. It found a seven-dimensional structure of ways of being independent and interdependent, with an important conclusion: the simple opposition “independent West—interdependent East” does not capture actual diversity. Groups can combine different forms of autonomy and relatedness (Vignoles et al., 2016).

India is probably the paradigmatic example of this combination. A person can be highly autonomous in a career yet highly interdependent within the family; highly conformist in one domain and innovative in another. Measuring personality without situational context risks missing precisely this contextual architecture.

What might a stratified society select?

In principle, institutions of endogamous marriage can produce genetic structure between subpopulations over the long term. But inferring selection on psychological traits from this is an enormous additional step, requiring specific genetic, historical, and reproductive data. It is more defensible to say that kinship networks, marriage, status, and family norms strongly modify cultural transmission and reproductive opportunities.

A highly stratified society can preserve several strategies simultaneously. In some groups, high female education may reduce fertility; in others, extended networks may absorb parenthood's costs. Urbanization can dismantle or transform these mechanisms. Internal migration can psychologically select who reaches Bangalore or Mumbai before the city has any effect on the individual.

From this book's perspective, India is not a case for a conclusion; it is a methodological warning. If you cannot describe the local institution, family structure, urban or rural environment, and cohort, you probably do not yet know which population you are comparing.

CHAPTER 10

The frontier, the island, and the migrant

Populations change because people move, not only because they are born.

Social selection often begins before reproduction: who leaves and who stays?

Migration is one of the most powerful psychological and demographic filters. Leaving a country, studying abroad, settling a frontier, or moving to a megacity involves costs, uncertainty, and broken relationships. It is unsurprising that mobility is not random with respect to personality.

In Zimmermann and Neyer's longitudinal study, initial extraversion and conscientiousness predicted participation in shorter-term international mobility, while extraversion and openness predicted longer-term mobility. After departure, personality changed too. A migrant population can therefore differ for two reasons: self-selection and socialization.

This mechanism complicates every urban-rural comparison. If more open or proactive young people move to cities, the difference between city and village is not created by the city alone. The city has filtered it. If those same young people then have fewer children, migration becomes a link between personality and fertility.

The American frontier

Research on frontier geographies and mountainous regions of the United States has found associations between environment, settlement history, and personality profiles. A study of more than 3.3 million people reported that mountain residents had, on average, greater openness and lower levels of traits such as agreeableness and extraversion, interpreted through a “frontier settlement theory.” The finding is observational and does not perfectly separate migration selection from socialization or historical persistence. But it shows how geography can retain psychological signatures through the flows of people who choose to inhabit it.

The island is not merely isolation

Islands and isolated communities are often imagined as genetic laboratories. Sometimes they are, particularly for Mendelian diseases or founder effects. For personality and behavior, however, social isolation matters as much as genetic isolation. A small community has dense reputational networks, few occupational niches, and high costs of deviance. Even without any genetic difference, these conditions can compress the observed distribution of behavior.

Here the niche diversity hypothesis returns: complex societies can sustain combinations of traits that would be difficult to turn into viable roles in a small community. When the number of niches grows, it is not only the economy that diversifies. Socially acceptable ways of being a person diversify too.

This forces us to reformulate future scenarios. A “neo-tribal” world would not necessarily genetically select conformity; it could, much more quickly, reduce the niches in which nonconforming behavior is sustainable. A world of megacities may preserve professional eccentricity while reproductively penalizing precisely the mobility and individualism that sustain it.

CHAPTER 11

When five factors become two

Social complexity can be a machine for preserving differences.

The Tsimane finding is easily misinterpreted. If the Big Five does not emerge robustly in a community of forager-horticulturalists, we might conclude that “traditional people have simpler personalities.” That would confuse the statistical model with the people.

A more interesting explanation is that the environment demands more correlated packages of behavior. In a subsistence society, the same person must work, cooperate with relatives, maintain alliances, raise children, and manage conflicts within a community where roles overlap. Being very industrious but completely unreliable socially may be harder to fit into a “niche” than in an economy with thousands of occupations.

Smaldino and colleagues' model starts here. More niches mean that more combinations of traits can yield satisfactory outcomes. If society has room for the withdrawn programmer, disciplined surgeon, extraverted actor, obsessive researcher, empathetic educator, and risk-tolerant entrepreneur, correlations between traits decrease. Statistical factors separate.

A subsequent study across 115 nations found precisely an association between niche diversity and a more differentiated personality structure. This is not definitive proof that niches cause factors, but it is strong convergence between model and data.

Selection or protection?

This idea reverses one of the book's initial intuitions. We were asking whether large bureaucratic societies gradually eliminate certain traits. Sometimes, on the contrary, they may protect variation through specialization. Large markets and complex states make profiles viable that would find no role in a village.

But economic protection does not guarantee reproduction. A profile can be highly viable professionally and fare very poorly reproductively. This creates the possibility of two-stage selection: complexity increases the diversity of adult lives, while family arrangements may make some of those lives almost incompatible with two or three children.

Here, perhaps, lies modernity's central tension. We have created history's richest ecosystem of identities, professions, and lifestyles without making all these niches compatible with reproduction. If this incompatibility is systematic, today's cultural pluralism may become tomorrow's narrower demographic composition.

The good society as an ecosystem of niches

From this perspective, freedom means more than the formal right to choose. It means the existence of viable niches for different choices. A society that claims to value research, art, care, entrepreneurship, and family, yet makes their combinations logistically impossible, has declarative pluralism and a much narrower practical selection.

This offers a new criterion for social design: how many kinds of people can live well and, if they wish, have children without disproportionate costs?

CENTRAL INSIGHT PLAUSIBLE AND TESTABLE

Social complexity may increase observable psychological diversity by multiplying niches. But if only some niches are compatible with family life, the diversity of adults and that of the next generation may move in different directions.

PART III

THE DARKER EDGES

Psychopathy, paranoia, and mental disorders are where evolutionary speculation must be tethered most closely to data.

CHAPTER 12

Psychopathy: predator or measurement problem?

When a dramatic word encompasses very different instruments and populations.

Few concepts attract evolutionary theories more readily than psychopathy. The image is intuitive: within a cooperative population, a cold, manipulative minority without remorse can exploit others' trust. If the strategy confers reproductive advantages, might society maintain or even amplify such traits?

Before the story, we need measurement. A 2021 meta-analysis of 15 studies, 16 samples, and 11,497 adults estimated a pooled prevalence of psychopathy in the general population of 4.5%. But when the definition used the PCL-R, the clinical-forensic benchmark instrument, the estimate was only 1.2%. Prevalence varied strongly by instrument, sex, and sample type (Sanz-García et al., 2021).

This difference is enormous. It tells us that “how many psychopaths are there?” cannot be answered without asking “how do we define them?” Self-report questionnaires of psychopathic traits and clinical assessment are not the same thing.

A frequency-dependent strategy?

Evolutionary models have sometimes proposed psychopathy as a frequency-dependent social strategy: exploitation works better when most people cooperate. This is a coherent hypothesis. But turning it into a claim about modern selection requires demonstrating three things: that the relevant traits are sufficiently heritable; that they increase reproductive success, not merely income, status, or number of partners; and that the association persists after accounting for social and environmental factors.

The data are much more modest. A recent study in a representative Croatian sample of 690 people found, among certain “Dark Tetrad” traits, a positive association between psychopathy and number of children, while sadism showed the opposite relationship. This is a provocative finding, not an evolutionary law. It comes from a single sample, measures traits, and lacks cross-cultural replication.

Institutions can select behavior without selecting the gene

It is more defensible to observe that certain institutions may temporarily reward behaviors resembling some components of psychopathy: risk tolerance, lack of anxiety, instrumental aggression, and the ability to dismiss employees or negotiate without emotional cost. In an environment with weak feedback and diffuse responsibility, the local advantage may grow.

But these components are not equivalent to clinical psychopathy. A calm surgeon, a commander able to decide under pressure, or a risk-tolerant entrepreneur is not, by virtue of these traits, a psychopath. The label can turn functional differences into pathology.

More importantly, institutions can be designed against exploitation: transparency, auditing, separation of powers, reputation, whistleblower protection, peer review. If the exploitative strategy depends on opacity, institutional architecture changes its cultural fitness.

The right question for the future is not whether “psychopaths will take over society,” but how far a large society can rebuild the feedback and reputational mechanisms that limit exploitation in small groups.

CHAPTER 13

Paranoia in the crowd

A mind built for local threats in a world of millions of strangers.

Paranoia is another concept that everyday language turns into a binary category. In research, persecutory thinking appears more as a continuum. A classic review estimated that 10–15% of the general population experiences paranoid thoughts relatively regularly, while severe persecutory delusions occur at the clinical end of the distribution (Freeman, 2007).

In a representative survey in England, broader expressions of suspicion were reported by far more people than severe ideas of personal conspiracy. This distribution is intuitive: cognitive systems for detecting hostile intentions do not start from zero and then suddenly become “illness.” They are normal mechanisms that stress, trauma, isolation, discrimination, or uncertainty can overcalibrate.

The city and the stranger

In a village, strangers are rare and reputation is public. In a city, we encounter hundreds of people every day about whom we know almost nothing. Institutions replace reputation with rules, cameras, contracts, online ratings, and documents. This world is simultaneously safer and more impersonal.

Without reducing illness to adaptation, paranoia can be viewed as a problem of threat-detection calibration. If the cost of missing an attack is high, a cognitive system may tolerate many false positives. But modern environments contain an immense flow of ambiguous signals: social media, global news, anonymous comments, algorithms maximizing engagement. A mechanism calibrated for small groups may receive too much unfiltered information.

This is a mismatch hypothesis, not proof that urbanization “selects paranoia.” Epidemiology shows associations between urbanicity and certain psychotic disorders, but possible causes include poverty, segregation, migration, discrimination, environmental exposures, service access, and residential selection.

Political paranoia and clinical paranoia

Large societies also create rational suspicions. Opaque institutions really can conspire; platforms really do collect data; influence campaigns really do exist. Calling every distrust “paranoia” can pathologize civic vigilance. Conversely, the reality of some conspiracies does not validate every conspiratorial belief.

For society's future, the important question is whether we can build infrastructures of verifiable trust. Transparency, auditability, and procedures for challenging decisions reduce the need to choose between credulity and total suspicion. In a society of strangers, trust cannot be emotion alone; it must partly be architecture.

CHAPTER 14

The city as a psychological experiment

The megacity is not merely a place. It is a new kind of human environment.

More than half of humanity lives in urban areas, and the proportion keeps growing. Evolutionarily, the city is a recent invention. Psychologically, it is an experiment in scale: high density, extensive networks, contact with strangers, noise, pollution, anonymity, opportunities, specialization, and mobility.

A meta-analysis of psychotic disorder incidence, based on 33 reports, found an incidence rate approximately 64% higher in urban than rural areas, with a confidence interval of 1.38–1.95. The same study found even larger effects for migrant status and socioeconomically disadvantaged areas (Castillejos et al., 2018). These associations matter, but they must not become a verdict that "the city causes psychosis." Urbanicity is a package of exposures and selection processes.

Other meta-analyses find associations between urban living and depression in developed countries. Loneliness and social isolation, although not exclusively urban, can paradoxically coexist with density. You can be surrounded by millions of people and lack repeated relationships.

The city also offers protection

The same density produces hospitals, universities, minority communities, specialized labor markets, and access to social partners unavailable in a village. For people with rare interests or unusual identities, the city can be liberating. The niche hypothesis predicts precisely this: complexity protects variety.

The comparison “healthy village—sick city” is therefore wrong. The village offers integration and support but may impose conformity, lack of anonymity, and few exits from local conflicts. The city offers autonomy and options but may fragment relationships and raise living costs.

The double effect: mind and fertility

What makes the city special for our subject is the overlap of two patterns: lower fertility and certain higher mental health risks. This does not mean the second causes the first. Together, however, they suggest that the modern megacity can be economically and culturally highly productive without being optimal for communities' biological and psychological reproduction.

If future cities house an even larger proportion of the population, their design becomes a demographic issue. Sufficiently large homes, short distances, access to nature, meeting places, schools, care services, and stable neighborhoods are not “amenities.” They may be the infrastructure that makes density compatible with family life and mental health.

The megacity need not be abandoned. It needs to be domesticated for the species that inhabits it.

CHAPTER 15

Mental disorders and reproduction

Genetic persistence does not imply reproductive advantage.

Here arises one of the most persistent evolutionary confusions: if a mental disorder is partly heritable and still exists in a population, it must once have conferred an advantage. Not necessarily.

Selection can maintain variants with mixed effects; mutation can reintroduce harmful variants; effects may occur after reproduction; the same variants can have different consequences in different combinations; polygenic architecture can prevent simple "purging." Persistence is not proof of adaptation.

A genomic study of 150,656 Icelanders without the diagnoses under analysis examined polygenic scores for autism, ADHD, schizophrenia, bipolar disorder, and major depression. A higher autism risk score was associated with fewer children and an older age at first birth; the ADHD score was associated with more children. The authors found no reproductive advantage for high schizophrenia or bipolar scores. Rare copy-number variants carrying moderate or high psychiatric risk were associated with fewer children (Mullins et al., 2017).

This picture is almost the opposite of the popular story that "illness spreads because modernity allows it." For many severe disorders, fertility is lower.

Norwegian population data published in 2025, based on national registers, generally find lower birth rates among people with mental disorders, with important differences between diagnoses and sexes. The relationships weaken when partnership, education, income, and other factors are considered, showing how extensively reproduction is mediated through social integration (Kravdal, Flatø & Torvik, 2025).

Why do they not disappear?

For polygenic traits and disorders, there is no single genetic switch for selection to turn off. Individual variants can have tiny and pleiotropic effects. Some components of a predisposition may be advantageous at subclinical levels and disadvantageous at extremes, but this must be demonstrated for each trait, not assumed.

ADHD is an interesting example precisely because its polygenic score was associated with more children in Iceland. But this does not mean that "ADHD is positively selected" as a diagnosis. The genetic score reflects a distribution of variants, the effects are small, and reproductive success within one society is not universal.

Policy without eugenics

If society affects reproduction differently for people with mental disorders, the ethical response is not to optimize genetic composition. It is to reduce suffering, discrimination, and barriers that turn a treatable condition into complete social exclusion. Reproductive freedom also includes a real possibility for people with disabilities or disorders to form relationships and families if they wish.

A humane system should be judged not by how efficiently it selects "good" people, but by how little it turns vulnerabilities into inherited social sentences.

CONCLUSION AGAINST SENSATIONALISM STRONG FOR THE GENERAL DIRECTION

Available evidence does not support the idea that modern societies produce a general increase in predispositions to severe mental disorders through reproductive advantage. For many diagnoses, the observed association runs in the opposite direction.

PART IV

FOUR FUTURES

Scenarios are not predictions. They are thought experiments built from observable mechanisms.

CHAPTER 16

The meritocratic megacity

A civilization that attracts the most mobile people, specializes them, and then makes them difficult to replace.

Let us extrapolate current trends. In 2100, much economic and cultural activity is concentrated in urban megaregions. Education takes longer. Competition for elite positions starts early and ends late. Central housing is expensive, and people frequently move in pursuit of opportunities. AI automates routine work but increases the value of those who coordinate complex systems, conduct research, create, or make high-stakes decisions.

This system is extraordinarily productive. It is also a potential reproductive filter.

Who enters? Probably a disproportionate number of mobile, educated, opportunity-oriented people, often more open to experience and more proactive. Who stays? Those who find good professional niches or communities. Who has three children? Perhaps not those with the greatest professional productivity, but those whose values, incomes, networks, or religions make large families compatible with the city.

Within a few generations, the megacity may become a cultural machine fueled from outside. It attracts people, changes their norms, reduces their fertility, and demands a new cohort of migrants. As long as a demographic hinterland exists, the system works. If fertility declines globally, competition for young people becomes geopolitical.

“Idiocracy” does not follow automatically

It is tempting to infer that if education correlates negatively with fertility, cognitive capital will inevitably decline. This conclusion is too simple. Education is not intelligence, polygenic scores have limited predictive power and are contextual, and the educational environment can offset or amplify differences. Furthermore, fertility by educational level can follow a nonlinear pattern and respond to policy.

The real risk is institutional: if society makes reproduction incompatible with precisely the paths it declares valuable, it creates tension between cultural and biological reproduction. Universities, laboratories, hospitals, and firms can continue to train highly capable people, but they depend on a population that wants and is able to produce the next cohort.

Selection for resilience to the city

In a persistently low-fertility megacity, long-term selection might emerge for combinations of predispositions that make reproduction more likely in that environment: a stronger preference for family, the ability to maintain stable partnerships, tolerance for crowding, dense religious or community networks. But cultural selection will almost certainly precede genetic selection. Groups with robust family norms can expand demographically much faster than a complex trait can evolve.

The meritocratic megacity does not select a single “urban human.” It first selects communities capable of reproducing under its conditions.

CHAPTER 17

The neo-tribal archipelago

More belonging, potentially more fertility—and a cost in tolerance between groups.

The second future is not a literal return to the village. It is the fragmentation of the world into dense identity communities: religious, ideological, ethnic, professional, or digital, some territorial and others globally distributed. AI makes small institutions administratively feasible, while remote work reduces the need for a metropolitan center. People choose communities with clearer rules and more repeated relationships.

Such a system can solve two of the megacity's problems: loneliness and the cost of family cooperation. Communities can share childcare, create dense reputational networks, and confer status on parenthood. If norms are pronatalist, fertility differences can become large.

From the perspective of cultural group selection, groups that sustain cooperation, transmit norms, and produce the next generation may grow relative to groups that do not. Culture can act here as a demographic multiplier.

The price of tightness

But dense community brings control. Strong norms reduce uncertainty and free-riding but may penalize deviance. People whose psychological profiles or values do not fit have fewer niches. If leaving the group is costly, protection becomes captivity.

More seriously, archipelagos may cooperate poorly with one another. High internal trust can coexist with suspicion toward outgroups. Communication algorithms can reinforce boundaries. Political paranoia may become functional for leaders who maintain cohesion through external threats.

Selection through differential fertility

If some communities sustain 3–4 children per woman and others 1–1.5 over many generations, their relative proportions change rapidly. What is transmitted is primarily culture: religion, norms, networks. If the relevant psychological traits are partly heritable and assortative mating exists, a genetic component may also emerge. But we should not assume it; the cultural effect is strong enough to explain much of the dynamic.

The neo-tribal archipelago is demographically robust but politically fragile. It maximizes local belonging while risking the loss of shared space.

The design problem is how to obtain the tribe's benefits without its total sovereignty: strong intermediate communities with a real right to leave; local identities with universal legal institutions; support networks without a monopoly over information and partners.

CHAPTER 18

The metropolis of proximity

Can density exist without atomization, and freedom without structural infertility?

The third scenario is less spectacular and probably more desirable: we do not abandon cities but reshape them around human social biology.

The metropolis of proximity is dense but does not force everyday life across long distances. Family housing is not exiled to the outskirts. Schools, parks, care services, healthcare, and some work are accessible nearby. Neighborhoods are stable enough for repeated relationships to develop, yet open enough for people to choose communities.

This is not merely urban planning. It is an intervention in the reproductive filters identified earlier.

It reduces the cost of the marginal child

If a second child does not require another hour of commuting, moving away from friends, and one parent abandoning a career, desired fertility is more likely to become realized fertility. From this perspective, effective family policy may concern time and space more than bonuses.

It rebuilds alloparenting

We cannot automatically recreate the clan, but we can build infrastructures in which parents are not alone: good nurseries, schools with compatible schedules, shared spaces, care cooperatives, optional intergenerational housing, flexible working hours. A two-income society can support family life if it does not ask two adults to simulate an entire village by themselves.

It preserves niches

Unlike neo-tribalism, the metropolis of proximity preserves role diversity. Unlike the hypercentralized megacity, it reduces the cost of everyday mobility and can stabilize local relationships. The ideal is a system with many professional niches and sufficiently small communities of everyday life.

This scenario also has an evolutionary virtue: it reduces artificial selection produced by incompatibility between family life and certain careers or personalities. It does not select a profile; it makes the reproductive environment more neutral. People who want children face fewer arbitrary obstacles, whether they are researchers, artists, entrepreneurs, introverts, extraverts, mobile, or locally rooted.

A philosophical norm: reasonable reproductive neutrality

Complete neutrality is impossible. Every society makes some lives easier than others. But we can pursue a criterion: institutions should not systematically turn socially valuable traits or legitimate lifestyles into a severe reproductive disadvantage without a necessary reason.

In this sense, demographic policy should not tell people how many children to have. It should reduce the gap between how many they say they want and how many they can realistically have.

A free society does not maximize birth rates. It maximizes the possibility that reproductive choices are choices, not accidents of infrastructure.

THE BOOK'S POSITION NORMATIVE RECOMMENDATION

Among the scenarios examined, the most robust direction is a polycentric society: dense yet family-compatible cities; strong but permeable local communities; many professional and cultural niches; a right to leave; infrastructure that reduces the artificial conflict between education, autonomy, and family life.

CHAPTER 19

Algorithmic reproduction

When selection comes not only from institutions but from the ranking function.

The fourth future is already partly here. Partners meet through platforms. Algorithms decide who becomes visible. Social reputation is filtered through scores, recommendations, and networks. Reproductive technologies increasingly allow sex, partnership, pregnancy, and genetic inheritance to be separated into distinct stages.

This world introduces a new form of selection: not necessarily genetic in the classical sense, but algorithmic before reproduction.

Matching changes the distribution of encounters

If a system recommends partners based on education, income, attractiveness, or preferences, it can increase assortative mating. Even if every user decides freely, the architecture of visibility changes probabilities. A ranking parameter can have aggregate demographic effects without having been designed for that purpose.

Platforms may also produce winner-take-more dynamics: a small share of users receives disproportionately large amounts of attention. If this pattern translates into relationships and reproduction, variance in mating success may increase. If not, it remains merely an attention economy. We need long-term data, not anecdotes.

From matching to explicit selection

Genetic screening for monogenic diseases is already a clinical reality. As polygenic scores become available for complex traits, the temptation emerges to optimize embryos for statistical probabilities. This requires radical caution. Scores are population-dependent, have limited predictive power for many traits, involve pleiotropic effects, and carry substantial uncertainties. Turning a probabilistic prediction into a hierarchy of “human quality” would be scientifically naive and ethically dangerous.

More subtle is market eugenics without a eugenicist. Nobody issues a decree; millions of families choose independently on the basis of the same commercial values and scores. If an interface presents certain characteristics as “better,” the system can standardize preferences on a large scale.

AI as a mediator of reproductive values

An AI that recommends partners or explains genetic risks is not neutral merely because it has no intentions. The optimization function, dataset, and framing of options encode values. In such a sensitive domain, diversity must be treated as a safety property.

Paradoxically, AI can also work in the opposite direction: reducing friction in family life, coordinating services, helping isolated people find compatible communities, making work flexible, and reducing parenthood's bureaucratic costs. The same technology can intensify filtering or neutralize arbitrary filters.

The future of algorithmic reproduction depends not only on what AI can calculate. It depends on what we refuse to turn into a score.

CHAPTER 20

The society we choose to select

We cannot eliminate selection. We can choose what kind of environment we build.

At this point, the opening question should look different. Not: “what type of person does society select?” But: “which institutional mechanisms make some life strategies reproductively viable and others improbable, and are these differences necessary or accidental?”

This reformulation changes politics too. If we want a healthy population and a pluralistic civilization, we do not need to choose genes. We need to design compatibilities.

1. Separate success from compulsory family sacrifice

Elite institutions often have a hidden norm: total availability precisely during the years when people form families. Grant programs, residencies, firms, universities, and administrations can shift rewards so that parenthood is not interpreted as a lack of seriousness. This is not merely equity; it reduces an arbitrary reproductive filter.

2. Protect niche diversity

A resilient society needs explorers and conservatives, introverts and extraverts, rule-oriented people and people who question rules. No optimal distribution can be calculated in advance for an uncertain future. Behavioral diversity functions as a portfolio of strategies.

3. Build intermediate communities

The nuclear family and the national state leave a gap between two adults and millions of citizens. Associations, neighborhoods, clubs, professional communities, cooperatives, schools, and friendship networks can fill this level. They reduce loneliness and rebuild cooperation without demanding total tribalism.

4. Preserve the right to leave

Strong communities become dangerous when the cost of leaving is prohibitive. A pluralistic society must permit dense belonging and mobility. This combination is difficult but essential to avoiding cultural selection through coercion.

5. Measure desired versus realized fertility

A low birth rate can reflect genuine preferences or barriers. Legitimate policy should not maximize births but understand the difference. If people want one child and have one, the problem is not the same as when they want two but have one because of housing, time, infertility, or lack of a partner.

6. Do not turn diagnosis into destiny

People with mental disorders or disabilities are not society's “genetic material.” Justified intervention addresses suffering and autonomy, not population purification. A decent civilization accepts that vulnerability is part of the human distribution.

7. Audit algorithms that mediate mating

Dating platforms, social networks, and potential reproductive tools can change who meets whom. Effects involving concentration, discrimination, and assortative mating deserve to be treated as social impacts, not merely product metrics.

8. Think in generations, not quarters

A policy can increase GDP while simultaneously reducing the capacity for social reproduction: time for children, housing stability, good schools, mental health, trust. Economic indicators do not automatically capture these externalities.

Every society is a theory of humanity written in incentives. Every generation tests that theory through the lives it manages to continue.

We cannot build a world without selection. Every environment favors some behaviors. But we can build a world where selection is not an opaque accident of rents, commuting, algorithms, and prestige. We can make pluralism not merely a declared value but a demographic property.

EPILOGUE

Disappearance is not an event

Species disappear when the last individual dies. Societies disappear differently: when the last child no longer learns the language, when a profession no longer has apprentices, when a community no longer produces adults who want to continue it, when institutions remain standing but must constantly be sustained from outside because their own way of life does not reproduce.

There is nothing moral about fertility as a number. A childless life can be complete, valuable, and freely chosen. A decent society does not turn reproduction into an obligation. But in aggregate, a civilization that does not produce a next generation cannot avoid the consequence by using more elegant vocabulary.

The great question of the twenty-first century is not whether humans will cease to exist. It is which forms of social life will prove compatible with continuity.

Perhaps megacities will learn to become overlapping villages. Perhaps AI will reduce the logistical cost of family life. Perhaps work will lose its centrality, and time will return to households and communities. Perhaps new tribes will offer belonging but need to be tempered by universal rights. Perhaps reproductive technologies will expand freedom without standardizing humanity. Or perhaps we will keep optimizing each institution locally—the university for performance, the city for land prices, the firm for profit, the platform for engagement—and discover that the global system has accidentally optimized against its own reproduction.

Selection has no plan. We do.

This is the moral difference between evolution and civilization. Evolution preserves what reproduces. Civilization can ask, beforehand, what deserves the possibility of continuing.

Evidence map and falsification criteria

The book has used three levels of certainty: robust and replicated findings; suggestive findings that depend on measurement and context; and speculative scenarios. The table below summarizes the claims supporting the central argument.

Claim	Confidence	Empirical basis	What would falsify or limit it
Personality has a moderate heritable component within populations.	High	Behavioral genetic meta-analysis; average h^2 ~0.40.	It would be weakened by larger meta-analyses showing near-zero estimates after controlling for biases.
Comparisons of mean personality levels across cultures are methodologically fragile.	High	Reviews of measurement invariance; lack of full scalar invariance.	Instruments with robust scalar invariance across many populations would allow stronger comparisons.

Claim	Confidence	Empirical basis	What would falsify or limit it
The Big Five structure is not reproduced identically in all societies.	High for the existence of exceptions	Tsimane; Karnataka sample; non-Western studies.	Emic and etic replications consistently recovering the same structure would narrow the conclusion.
Social niche diversity is associated with more statistically differentiated personality.	Moderate to high	Formal model plus cross-national analyses covering up to 115 countries.	Longitudinal or quasi-experimental studies finding no changes when niches diversify.
Fertility is generally lower in urban than rural Europe.	High	Eurostat and multinational regional studies.	A persistent reversal of the gradient in most countries after controlling for composition.
Some personality traits covary with number of children.	Moderate	Multicountry and cohort studies; small, heterogeneous effects.	Preregistered meta-analyses finding stable null effects for all traits.

Claim	Confidence	Empirical basis	What would falsify or limit it
Modern genetic selection on complex traits can be detected.	Moderate to high	Polygenic scores and reproductive data from Iceland/UK; reproductive success loci.	If associations systematically disappear within families and independent samples, or are fully explained by stratification.
Society genetically selects a universal psychological profile.	Low / unsupported	Gradients differ across countries; cross-cultural measurement is fragile.	This would require cross-culturally replicable, heritable effects stable over many generations.
Psychopathy affects ~1% or ~4.5% of the general population.	Definition-dependent	1.2% using PCL-R; 4.5% pooled across diverse instruments.	A consensus instrument and threshold, applied representativel y worldwide, would narrow the range.

Claim	Confidence	Empirical basis	What would falsify or limit it
Urbanicity is associated with higher psychosis incidence.	Moderate to high	Meta-analysis: urban/rural IRR ~1.64.	Natural experiments and causal models eliminating the association after accounting for confounders.
Severe mental disorders tend to reduce fertility.	High for several diagnoses	Nordic registers; Icelandic genetic data.	Representative cohorts in different contexts showing equal or higher fertility after adjustments.
A metropolis of proximity may reduce arbitrary reproductive filtering.	Speculative, testable	Derived from mechanisms involving costs, time, housing, and support.	Urban planning or family interventions producing no change in realized fertility or parental well-being.

MINIMAL GLOSSARY

Assortative mating: Nonrandom pairing: partners resemble each other more than chance would predict in education, values, traits, or background.

Reproductive fitness: Contribution to future generations, often approximated by the number of children or descendants who reproduce. It is not synonymous with health, moral worth, or social success.

Heritability: The proportion of variation in a trait within a population and environment statistically attributed to genetic differences. It is not a measure of individual determinism.

Measurement invariance: The condition that a psychological instrument measures the same construct comparably across groups. Scalar invariance is important for comparisons of means.

Socioecological niche: A role or combination of social, economic, and ecological demands within which a particular behavioral profile can function.

Polygenic score: A statistical aggregation of many genetic variants associated with an outcome in a GWAS. It has limited predictive power and depends on the reference population.

Cultural selection: The differential spread of ideas, norms, and practices through imitation, prestige, institutions, sanctions, and transmission between generations.

Genetic selection: Changes in genetic frequencies produced by the association between heritable variation and reproductive success.

Tightness-looseness: The degree to which a culture has strong norms and low versus high tolerance for deviance.

SELECTED BIBLIOGRAPHY

Sources were selected for the book's central claims. For controversial subjects, preference was given to meta-analyses, population registers, large-sample publications, and articles explicitly criticizing popular findings.

Castillejos, M. C. et al. (2018). A systematic review and meta-analysis of the incidence of psychotic disorders: the distribution of rates and the influence of gender, urbanicity, immigration and socio-economic level. Psychological Medicine. https://pubmed.ncbi.nlm.nih.gov/29467052/

Cheung, F. M. et al. (2003). Indigenous measures of personality assessment in Asian countries: a review. Psychological Assessment, 15(3), 280-289. DOI: 10.1037/1040-3590.15.3.280. https://pubmed.ncbi.nlm.nih.gov/14593828/

Eurostat. (2024). Demography of Europe - 2024 edition. Urban-rural fertility statistics. https://ec.europa.eu/eurostat/web/interactive-publications/demography-2024

Freeman, D. (2007). Suspicious minds: the psychology of persecutory delusions. Clinical Psychology Review, 27(4), 425-457. DOI: 10.1016/j.cpr.2006.10.004. https://pubmed.ncbi.nlm.nih.gov/17258852/

GBD 2021 Fertility and Forecasting Collaborators. (2024). Global fertility in 204 countries and territories, 1950-2021, with forecasts to 2100. The Lancet, 403, 2057-2099. DOI: 10.1016/S0140-6736(24)00550-6. https://pubmed.ncbi.nlm.nih.gov/38521087/

Gelfand, M. J. et al. (2011). Differences between tight and loose cultures: a 33-nation study. Science, 332, 1100-1104. https://pubmed.ncbi.nlm.nih.gov/21617077/

Gurven, M., von Rueden, C., Massenkoff, M., Kaplan, H., & Lero Vie, M. (2013). How universal is the Big Five? Testing the five-factor model among forager-farmers in the Bolivian Amazon. Journal of Personality and Social Psychology, 104(2), 354-370. DOI: 10.1037/a0030841. https://pmc.ncbi.nlm.nih.gov/articles/PMC4104167/

Henrich, J., Heine, S. J., & Norenzayan, A. (2010). The weirdest people in the world? Behavioral and Brain Sciences, 33, 61-83. DOI: 10.1017/S0140525X0999152X. https://pubmed.ncbi.nlm.nih.gov/20550733/

Ji, X., Liu, Z., & Zhu, T. (2026). Clan ties, collective minds: the persistent impact of Chinese kinship structures on modern collectivism. Humanities and Social Sciences Communications, 13, 724. https://www.nature.com/articles/s41599-026-07079-1

Kong, A. et al. (2017). Selection against variants in the genome associated with educational attainment. PNAS, 114(5), E727-E732. DOI: 10.1073/pnas.1612113114. https://pubmed.ncbi.nlm.nih.gov/28096410/

Kravdal, Ø., Flatø, M., & Torvik, F. A. (2025). Fertility among Norwegian Women and Men with Mental Disorders. European Journal of Population, 41, 17. DOI: 10.1007/s10680-025-09739-5. https://link.springer.com/article/10.1007/s10680-025-09739-5

Lukaszewski, A. W. et al. (2022). Niche Diversity Predicts Personality Structure Across 115 Nations. Psychological Science. https://pubmed.ncbi.nlm.nih.gov/35044268/

Mullins, N. et al. (2017). Reproductive fitness and genetic risk of psychiatric disorders in the general population. Nature Communications, 8. https://pmc.ncbi.nlm.nih.gov/articles/PMC5474730/

Riederer, B., & Beaujouan, É. (2024). Explaining the urban-rural gradient in later fertility in Europe. Population, Space and Place, 30(1), e2720. DOI: 10.1002/psp.2720. https://onlinelibrary.wiley.com/doi/10.1002/psp.2720

Ruan, J., Xie, Z., & Zhang, X. (2015). Does rice farming shape individualism and innovation? Food Policy, 56, 51-58. DOI: 10.1016/j.foodpol.2015.07.010. https://www.sciencedirect.com/science/article/pii/S0306919215000986

Sanz-García, A. et al. (2021). Prevalence of Psychopathy in the General Adult Population: A Systematic Review and Meta-Analysis. Frontiers in Psychology, 12. https://pmc.ncbi.nlm.nih.gov/articles/PMC8374040/

Sanz-García, A. et al. (2025). Consistency and Variation in Natural Selection on Personality Across 17 Countries. Personality and Social Psychology Bulletin / indexed PubMed. https://pubmed.ncbi.nlm.nih.gov/40094705/

Sheppard, D. et al. (2026). Understanding and assessing personality across cultures: A scoping review. PLOS ONE. DOI: 10.1371/journal.pone.0338521. https://pubmed.ncbi.nlm.nih.gov/41481748/

Smaldino, P. E., Lukaszewski, A., von Rueden, C., & Gurven, M. (2019). Niche diversity can explain cross-cultural differences in personality structure. Nature Human Behaviour, 3, 1276-1283. DOI: 10.1038/s41562-019-0730-3. https://www.nature.com/articles/s41562-019-0730-3

Talhelm, T. et al. (2014). Large-scale psychological differences within China explained by rice versus wheat agriculture. Science, 344(6184), 603-608. DOI: 10.1126/science.1246850. https://pubmed.ncbi.nlm.nih.gov/24812395/

Terracciano, A. et al. (2005). National character does not reflect mean personality trait levels in 49 cultures. Science, 310(5745), 96-100. DOI: 10.1126/science.1117199. https://pubmed.ncbi.nlm.nih.gov/16210536/

Thomas, S. et al. (2023). The fundamentals of Indian personality: An investigation of the big five. Indian Journal of Psychiatry / PMC. https://pmc.ncbi.nlm.nih.gov/articles/PMC10725210/

United Nations Department of Economic and Social Affairs, Population Division. (2024). World Population Prospects 2024: Summary of Results. https://www.un.org/development/desa/pd/content/world-population-prospects-2024

Vignoles, V. L. et al. (2016). Beyond the East-West dichotomy: Global variation in cultural models of selfhood. Journal of Experimental

Psychology: General, 145(8), 966-1000.
https://pubmed.ncbi.nlm.nih.gov/27359126/

Vukasović, T., & Bratko, D. (2015). Heritability of personality: A meta-analysis of behavior genetic studies. Psychological Bulletin, 141(4), 769-785. DOI: 10.1037/bul0000017.
https://pubmed.ncbi.nlm.nih.gov/25961374/

Zimmermann, J., & Neyer, F. J. (2013). Do we become a different person when hitting the road? Personality development of sojourners. Journal of Personality and Social Psychology, 105(3), 515-530. DOI: 10.1037/a0033019. https://pubmed.ncbi.nlm.nih.gov/23773042/

A society does not choose its descendants directly.
It chooses its rules, cities, sources of prestige, and costs.
Those choices then select, statistically, who can have a future.